# CMS Physics Analysis Summary

Contact: cms-pag-conveners-b2g@cern.ch 2016/07/17

# Search for diboson resonances in the semileptonic $X \to ZV \to \ell^+\ell^-\, q\bar{q}$ final state at $\sqrt{s} = 13$ TeV with CMS

The CMS Collaboration

## Abstract

A search for diboson resonances is performed in the semileptonic $X \to ZV \to \ell^+\ell^-\, q\bar{q}$ final state, where one vector boson decays hadronically ($V = W, Z$) and the other, a Z boson, decays into two leptons, in the mass range $550 \leq m_{ZV} \leq 2500$ GeV. In this regime the V boson may be sufficiently boosted that its decay products are contained in a single jet, thus merged jet reconstruction and substructure techniques are utilised. The case where the hadronic decay products are resolved is also considered at low mass. At high signal masses the closeness of the leptons is taken into account to avoid leptonic Z reconstruction inefficiencies. The analysis exploits the kinematic and flavour information of the final state particles of the event in order to isolate the hypothetical signal. The results are based on the analysis of proton-proton collision data collected by CMS in 2015, corresponding to an integrated luminosity of 2.7 $fb^{-1}$ delivered by the LHC at center-of-mass energy of 13 TeV.

# 1 Introduction

There are several theoretical models which motivate the existence of heavy particles that decay to pairs of bosons. These models usually address open questions of the standard model (SM) such as the breaking of the electroweak symmetry [1], or the integration of gravity into the SM using extra dimensions. Among the models under scrutiny in this analysis are the Warped Extra Dimension (WED) models [2, 3], which predict the existence of a spin 2 massive graviton. In the original Randall-Sundrum WED model (RS) only gravity is allowed to propagate in the extra dimensional bulk space; alternatively, in the bulk WED model [4–6] (bulk) both gravity and SM particles are allowed to propagate along the extra dimension.

This document reports the results of a search for diboson resonances in the semileptonic $\mathrm{X} \rightarrow \mathrm{ZV} \rightarrow \ell^+\ell^- \mathrm{q}\overline{\mathrm{q}}$ final state where one boson decays hadronically ($\mathrm{V} = \mathrm{W}, \mathrm{Z}$) and the other, a Z boson, decays to two leptons. The search uses 2.7 $\mathrm{fb}^{-1}$ of LHC proton-proton collision data at $\sqrt{s}$ = 13 TeV collected by CMS during the 2015 run and is performed in the mass range 550 $\leq m_{\mathrm{ZV}} \leq$ 2500 GeV. Within this broad mass range two specialised search strategies, focused on low and high mass signal searches, are defined.

In the mass regime of interest the hadronically decaying boson may be sufficiently boosted that its decay products are contained in a single merged jet; this is already a significant effect for signal masses around 600 GeV. If the hadronic decay products are resolved the boson decay products may be reconstructed as two jets (a dijet). Therefore dijet reconstruction is used in the low mass search strategy for events in which no boosted V candidates are found. At the experimental level, the RS and bulk WED models can be considered as two limiting cases in terms of the relative sensitivities of the merged jet and dijet analyses; in particular the $\Delta R(j, j)$ of the hadronically decaying bosons is larger within the RS model. For the high mass search strategy only boosted V reconstruction is required. Both search strategies make use of jet substructure techniques to determine the merged jet mass and identify merged jets with two well defined subjets, helping to discriminate vector bosons from QCD jets originating from quarks and gluons. Jet flavour categorisation is used to further increase the signal sensitivity. In the low mass search a categorisation based on the tagging of b jets is implemented, exploiting the relatively large branching fraction of signal boson decays to a pair of bottom-antibottom quarks compared to the abundance of light-quark or gluon jets in Z+jets background events. For the high mass regime a categorisation based on the substructure of boosted V jets is found to be more effective.

Leptonic Z decays begin to suffer from reconstruction inefficiencies due to the closeness of the leptons at signal masses around 1200 GeV. Therefore the high mass search strategy makes use of looser reconstruction criteria where appropiate and uses triggers that do not contain isolation requirements.

The low mass strategy searches for resonances in the mass range $550 \leq m_{\mathrm{ZV}} \leq 1400$ GeV; results are interpreted in the framework of the RS and bulk graviton models. The high mass strategy search explores the mass range $800 \leq m_{\mathrm{ZV}} \leq 2500$ GeV; results are interpreted in the context of the bulk graviton model. A boundary at 800 GeV is used to separate the results of the low and high mass bulk graviton search strategies. This separation point is motivated by the preservation of the bulk signal sensitivity over a large mass range; around 800 GeV the low mass strategy becomes less sensitive with respect to the high mass strategy. A summary of the analysis strategies adopted for the bulk signal search is given in Table 1.

This analysis largely follows the strategies of the Run 1 low mass $\mathrm{H} \rightarrow \mathrm{ZZ} \rightarrow \ell^+\ell^- \mathrm{q}\overline{\mathrm{q}}$ search [1] and the high mass diboson search [7] performed at $\sqrt{s} = 8$ TeV. Section 2 briefly describes the

Table 1: Analysis strategies for bulk graviton search.

| Strategy | Bulk signal search range | Final state | Categorisation |
|---|---|---|---|
| low mass | $550 \leq m_{ZV} \leq 1400$ GeV | $\ell\ell$ + V(1 or 2 jets) | Hadronic V type<br>b tagging criteria |
| high mass | $800 \leq m_{ZV} \leq 2500$ GeV | $\ell\ell$ (close by lepton effects)<br>+ V(1 jet) | V tagging criteria |

CMS detector; section 3 gives an overview of the simulations used in this analysis; section 4 provides a detailed description of the reconstruction and event selection; section 5 describes the background estimation. Systematic uncertainties are discussed in section 6. Finally, the results of the search for different graviton mass hypotheses are discussed in section 7.

## 2 The CMS Detector

The central feature of the CMS apparatus [8] is a superconducting solenoid of 6 m internal diameter, providing a magnetic field of 3.8 T. Contained within the solenoid volume are a silicon pixel and strip tracker, a lead tungstate crystal electromagnetic calorimeter (ECAL), and a brass and scintillator hadron calorimeter (HCAL), each composed of a barrel and two endcap sections. Muons are measured in gas-ionization detectors embedded in the steel flux-return yoke outside the solenoid. Extensive forward calorimetry complements the coverage provided by the barrel and endcap detectors.

The particle-flow (PF) event algorithm [9, 10] reconstructs and identifies individual particles with an optimized combination of information from all the CMS sub-detector systems. The reconstructed PF constituents are assigned to one of five candidate categories: electrons, muons, photons, charged and neutral hadrons. The energy of photons is directly obtained from the ECAL measurement, corrected for zero-suppression effects. The energy of electrons is determined from a combination of the electron momentum at the primary interaction vertex as determined by the tracker, the energy of the corresponding ECAL cluster, and the energy sum of all bremsstrahlung photons spatially compatible with originating from the electron track. The momentum of muons is obtained from the curvature of the corresponding track. The energy of charged hadrons is determined from a combination of their momentum measured in the tracker and the matching ECAL and HCAL energy deposits, corrected for zero-suppression effects and for the response function of the calorimeters to hadronic showers. Finally, the energy of neutral hadrons is obtained from the corresponding corrected ECAL and HCAL energies.

Hadronic jets are clustered from the four-momenta of the particles reconstructed by the PF algorithm, using the FastJet software package [11]. A correction based on the projected area of the jet on the front face of the calorimeter is used to take into account the extra energy clustered in jets coming from additional proton-proton interactions within the same bunch crossing (pileup). Jet energy corrections are derived from simulation and from dijet and photon+jet events in data as in ref. [12]. Additional quality criteria are applied to the jets in order to remove spurious jet-like features originating from isolated noise patterns in the calorimeters or the tracker. The efficiency of these jet quality requirements for signal events is above 99%.

Muons [13] are measured in the pseudorapidity range $|\eta| < 2.4$, with detection planes made using three technologies: drift tubes, cathode strip chambers, and resistive plate chambers. Matching muons to tracks measured in the silicon tracker results in a relative transverse momentum resolution for muons with $20 < p_T < 100$ GeV of 1.3–2.0% in the barrel and better than 6% in the endcaps. The $p_T$ resolution in the barrel is better than 10% for muons with $p_T$ up to

1 TeV.

Electrons [14] are measured in the electromagnetic calorimeter, in the pseudorapidity range $|\eta| < 2.5$. The electron momentum is estimated by combining the energy measurement in the ECAL with the momentum measurement in the tracker. The momentum resolution for electrons with $p_T \approx 45$ GeV from $Z \to ee$ decays ranges from 1.7% for nonshowering electrons in the barrel region to 4.5% for showering electrons in the endcaps.

A more detailed description of the CMS detector, together with a definition of the coordinate system used and the relevant kinematic variables, can be found in ref. [8].

# 3 Simulated Samples

For the description of the Z+jets background, an inclusive NLO sample and a larger set of exclusive LO samples are generated with the MADGRAPH5_AMC@NLO event generator [15]. The exclusive LO samples are binned according to HT (the sum of the hadron transverse energies at generator level). Samples of $t\bar{t}$ and SM diboson production are generated using POWHEG 2 [16–18] and PYTHIA 8 [19, 20], respectively. RS graviton signal events are generated using PYTHIA 8 with the ratio of the curvature parameter $k$ over the reduced Planck mass $M_{Pl}$ set to 0.1; bulk graviton signal events are generated with MADGRAPH 5 and $k/M_{Pl} = 0.5$. Under these conditions the natural width of the resonance is much smaller than the experimental resolution. Parton showering and hadronization processes are simulated by interfacing the event generators to PYTHIA 8 with the CUETP8M1 [21] tune. The NNPDF 3.0 parton distribution functions (PDF) [22] are used to model the momentum distribution of the colliding partons inside the protons. For both signal and background Monte Carlo (MC) samples, events are simulated using a GEANT4-based model [23] of the CMS detector and processed using the same reconstruction algorithms as for data.

Supplementary minimum bias interactions are added to the generated events in order to match the additional particle production observed in data from the large number of pileup interactions. The simulated samples are corrected for observed differences between data and simulation in the efficiencies of lepton trigger, lepton identification/isolation, and selection criteria identifying the flavour of the jets (b jets, boosted V jets).

# 4 Reconstruction and Selection

## 4.1 Trigger and Preliminary Offline Reconstruction

Events used in this analysis are recorded with an online selection algorithm (trigger) that requires a single electron or muon. For the low mass strategy, a triggering electron must have $p_T > 22$ GeV and $|\eta| < 2.1$, while a triggering muon must have $p_T > 20$ GeV; in both cases, the leptons are subject to loose identification and isolation requirements. For the high mass search, the triggering electron (muon) thresholds are raised to 105 (45) GeV and the isolation requirements dropped, avoiding efficiency loss due to close-by lepton effects.

In the subsequent selection, events are required to have at least one reconstructed collision vertex within the distance of 24 cm along the beam axis and 2 cm in the transverse plane to the beams from the nominal pp interaction point.

## 4.2 Lepton Reconstruction and Identification

For the low mass search, the leading and subleading leptons are required to have $p_T$ greater than 40 GeV and 24 GeV respectively. Muons (electrons) in the pseudorapidity range $|\eta| < 2.4$ ($|\eta| < 2.1$) are considered. Electrons in the transition region between the barrel and endcap, $1.44 < |\eta| < 1.57$, are excluded. The electron $\Delta R(ee)$ must be larger than 0.3. The $p_T(\ell)$, $\eta(\ell)$ and $\Delta R(ee)$ requirements are consistent with those in the online trigger algorithms. For the high mass search, the $p_T(\ell)$ requirements on the leading lepton are tighter with respect to the low mass case. The leading muon (electron) is required to have $p_T$ greater than 50 GeV (115 GeV), and pseudorapidity range $|\eta| < 2.1$ ($|\eta| < 2.5$); the subleading muon (electron) is required to have $p_T$ greater than 20 GeV (20 GeV), and pseudorapidity range $|\eta| < 2.4$ ($|\eta| < 2.5$). A looser muon selection is applied whereby one of the two muons of the leptonic Z candidate may be identified in the tracker system only.

Leptons are required to be isolated from hadronic activity in the event. The analysis uses charged and neutral PF particles in a cone of $\Delta R < 0.4$ (0.3) around the muon (electron) direction. The isolation sum $S^{\ell}_{\mathrm{Iso}}$ is defined as the scalar sum of the transverse momenta of charged hadrons originating from the primary vertex, plus the scalar sums of the transverse energies for neutral hadrons and photons, respectively. This sum is corrected to account for the contribution from neutrals originating from additional pp collisions in the same bunch crossing (pileup). For the high mass search a simpler muon isolation based only on reconstructed tracks is used. Here the $p_T$ of the inner track of one muon is removed from the isolation cone of the other, to avoid efficiency loss in the boosted regime.

Reconstruction efficiencies for leptons and their uncertainties are evaluated from data with a tag-and-probe technique [24], where one lepton from an inclusive sample of Z decays serves as a tag and the efficiency for the reconstruction of the other lepton is measured.

## 4.3 Hadronic V Reconstruction

Jets are clustered using the anti-$k_T$ algorithm [25] with a distance parameter $R$ equal either to 0.4 ("AK4 jets") or 0.8 ("AK8 jets"). Charged PF constituents not associated to the primary vertex are not used in the jet clustering procedure. Jets are required to be separated from isolated leptons by $\Delta R > 0.8$ (0.4) for AK8 (AK4) jets.

Depending on the transverse momentum of the V boson in the $\mathrm{V} \to \mathrm{q\bar{q}}$ process, the decay products may either be reconstructed as a single AK8 merged jet (boosted scenario) or as a pair of AK4 jets (resolved scenario). The low mass strategy addresses both of these cases, applying a categorisation based on the hadronic V reconstruction type; the high mass strategy focuses exclusively on the boosted scenario.

For AK8 jets with $p_T$ greater than 200 GeV a jet-pruning algorithm [26] is used to improve background rejection and jet mass resolution. Additionally, the n-subjettiness variable $\tau_{21}$ [27] is used to differentiate signal and background. The high purity (HP) category is defined by the requirement $\tau_{21} < 0.45$, and it is used for both the low and high mass searches. Additionally the high mass search defines the low purity (LP) category for events in the region $0.45 < \tau_{21} < 0.75$. For an AK8 jet to be considered V-tagged, the jet pruned mass is required to be in the signal region $65 < m_J < 105$ GeV. The V tagging efficiency, based on the requirement $65 < m_J < 105$ GeV and $\tau_{21} < 0.45$, is found to be 61% for a bulk graviton of 1 TeV and 23% for the simulated background. Corrections to the V tagging efficiency have been derived from data using a sample enriched in W hadronic decays from semileptonic $\mathrm{t\bar{t}}$ events [28].

Boosted V candidates with pruned masses in the ranges $30 < m_J < 65$ GeV and $135 < m_J < 180$

GeV, referred to as the sideband, are used to validate and normalize the simulation in the signal region of the low mass search. For the high mass search the sideband is in the ranges $20 < m_{\mathrm{J}} < 65$ GeV and $135 < m_{\mathrm{J}} < 220$ GeV and is used to estimate the background contribution in the signal region.

For events which do not pass the boosted V selection, the low mass search attempts to reconstruct a resolved hadronic V decay using two AK4 jets both with $p_{\mathrm{T}} > 30\,\mathrm{GeV}$. The resulting dijet system is required to have invariant mass $M_{jj} > 30\,\mathrm{GeV}$ and $p_{\mathrm{T}}(jj) > 100\,\mathrm{GeV}$. In events with more than two jets the best dijet combination is chosen based on the following criteria: the dijet candidate in the highest b-tag category is selected; if more than one candidate remains, the one closest in mass to the Z is chosen. The signal region for the low mass resolved category contains events where the dijet system has $m_{\mathrm{jj}}$ in the $65 < m_{\mathrm{jj}} < 110$ GeV range, whilst the sideband region used for background estimation is defined as for the low mass boosted category but with the upper mass threshold raised from 180 to 300 GeV to take advantage of the increased statistics in this region.

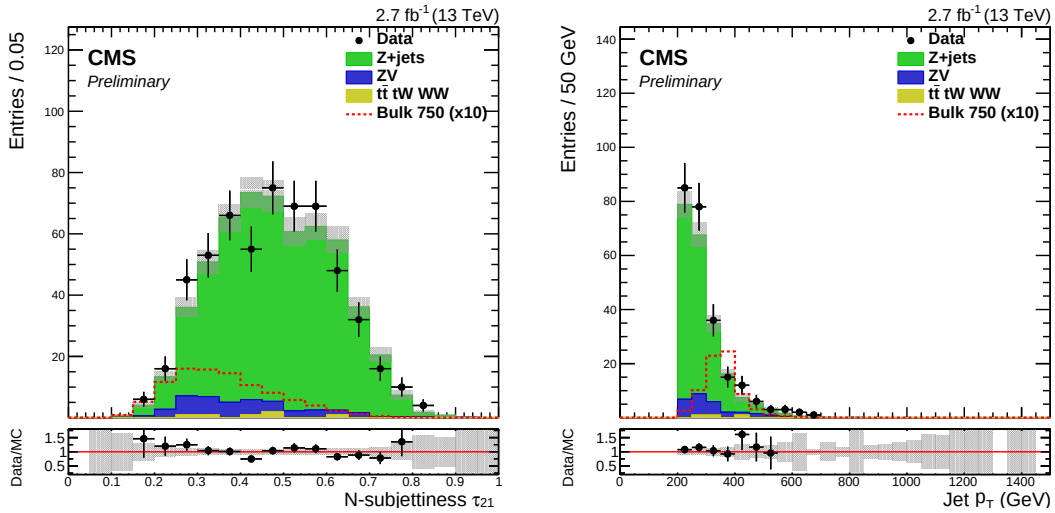


Figure 1: Distribution of the boosted V candidate $\tau_{21}$ and $p_{\mathrm{T}}(J)$ (after applying the $\tau_{21} < 0.45$ cut) in data and simulation for events in the signal region of the low mass search. Backgrounds are shown separately for the dominant Z+jets contribution (green), the ZV SM diboson contribution (blue), and the lepton flavour symmetric backgrounds derived from e$\mu$ data (yellow). The gray band shows the statistical and systematic uncertainties on the background. The background for the $\tau_2/\tau_1$ distribution is normalised to the number of events in the data; the background for the $p_{\mathrm{T}}(J)$ distribution is derived from a fit to the $m_{\mathrm{ZV}}$ observable in the sideband region alone.

The boosted V candidate $\tau_{21}$ and $p_{\mathrm{T}}(J)$ distributions for events in the low mass search are shown in Fig. 1. The $m_{\mathrm{J}}$ and $m_{\mathrm{jj}}$ distributions for events with a V candidate, boosted and resolved respectively, are shown in Fig. 2.

To identify jets originating from the hadronization of bottom quarks, the Combined Secondary Vertex (CSV) b tagging algorithm [29] is used, either directly on the AK4 jets (in the resolved category) or on the subjets of the AK8 jets (in the boosted category). The ratio between data and simulation for the b tagging and mistagging efficiencies are used as a scale factor to correct the simulated events. Distributions of the CSV b-tag discriminant are shown in Fig. 3.

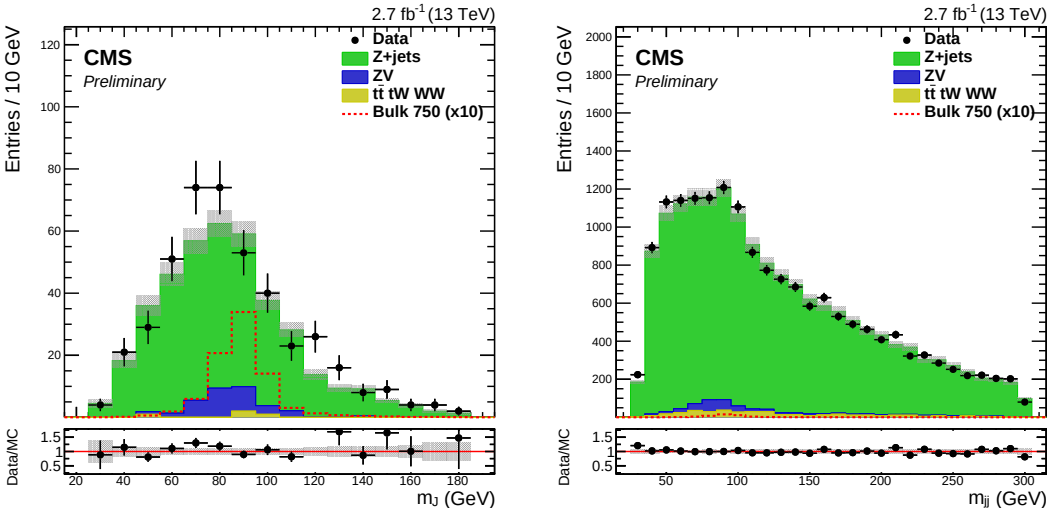


Figure 2: Distribution of V candidate $m_J$ (left) and $m_{jj}$ (right) in data and simulation for events in the signal and sideband regions of the low mass search. Backgrounds are shown seperately for the dominant Z+jets contribution (green), the ZV SM diboson contribution (blue), and the lepton flavour symmetric backgrounds derived from e$\mu$ data (yellow). The gray band shows the statistical and systematic uncertainties on the background. Background normalizations are derived from a fit to the $m_{ZV}$ observable in the sideband region alone.

## 4.4 Leptonic Z Reconstruction

Each pair of same flavour oppositely charged leptons is considered as a Z candidate. To suppress the combinatorial background from non Z lepton pairs, such as from $t\bar{t}$, while preserving a large acceptance to signal events the dilepton invariant mass $M_{\ell\ell}$ is required to be in the range $76 < M_{\ell\ell} < 106$ GeV in the low mass search, and $70 < M_{\ell\ell} < 110$ GeV in the high mass search.

## 4.5 Final Event Selection

Events are considered for further analysis if they contain at least one leptonic Z candidate and one hadronic V candidate. The diboson resonance candidate is formed by combining the hadronic V candidate and the leptonic Z candidate. The search is performed on the invariant mass distribution of the VZ candidates, $m_{ZV}$. In order to obtain the best possible sensitivity, events are classified according to lepton flavour, hadronic V type (boosted or resolved), b-tag criteria (tagged and untagged) and boosted V tagging criteria (low and high purity). The spectra of the reconstructed diboson mass, $m_{ZV}$, are used to set limits on or possibly discover a new resonance using a combined likelihood function. The categories are constructed to be exclusive.

For the low mass search, the events are first categorised by the hadronic V type: events with $p_T(\ell^+\ell^-) > 200$ GeV that contain an AK8 jet meeting the $p_T(J)$, $\tau_{21}$ and $m_J$ requirements discussed above constitute the boosted category. Of the remaining events, those with $p_T(\ell^+\ell^-) > 100$ GeV and containing two AK4 jets satisfying the $p_T(jj)$ and $m_{jj}$ requirements discussed above form the resolved category. Events failing this latter requirement are discarded. The boosted and resolved categories are further split into two exclusive b-tag categories. Events in the boosted tagged category are required to have one subjet identified with the medium ($\sim$65% efficiency) tag requirement (CSVM); events failing this requirement constitute the boosted untagged category. In the resolved case events in the tagged category are required to have one jet

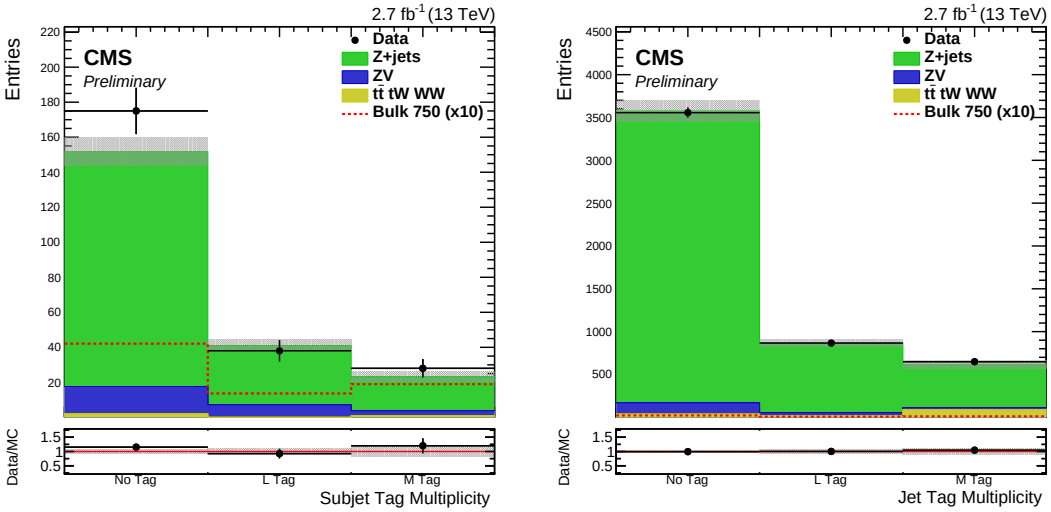


Figure 3: The zero, loose and medium CSV tag multiplicity for the most b-like subjet of the boosted V candidates (left) and the most b-like jet of the resolved V candidates (right) for events in the signal region of the low mass search. The gray band shows the statistical and systematic uncertainties on the background. Background normalizations are derived from a fit to the $m_{ZV}$ observable in the sideband region alone.

identified with the medium (∼65% efficiency and ∼1% light flavour mis-tag rate) tag requirement and the other jet with the loose (∼80% efficiency and ∼10% light flavour mistag rate) tag requirement (CSVL), with the remaining events constituting the resolved untagged category.

The high mass search addresses only the boosted V scenario, with a categorisation set by the n-subjettiness variable $\tau_{21}$. Fig. 4 shows the $\tau_{21}$ distribution for data and simulation, in the high and low purity categories. The high purity category is more efficient over the whole signal mass range, but suffers from large statistical uncertainties in data and simulation. On the other hand, the low purity category increases the expected signal yield and is more advantageous for the stability of the fits.

The final selection requirements and categorisation schemes are summarised in Tab. 2.

Table 2: Summary of selection requirements and categorisation.

| Observable | Selection criterion | | |
|---|---|---|---|
| | Low Mass Resolved | Low Mass Boosted | High Mass |
| | same lepton flavour, opposite charge | | |
| $p_T(\ell)$( GeV) | $> 40/24$ | $> 40/24$ | e($\mu$) $> 115/35(50/20)$ |
| $\lvert\eta(\ell)\rvert$ | (e) $< 2.1$, ($\mu$) $< 2.4$ | (e) $< 2.1$, ($\mu$) $< 2.4$ | (e) $< 2.5$, ($\mu$) $< 2.1/2.4$ |
| $\Delta R(\ell\ell)$ | (ee) $> 0.3$ | (ee) $> 0.3$ | – |
| $M(\ell^+\ell^-)$( GeV) | [76, 106] | [76, 106] | [70, 110] |
| $\lvert\eta(j)\rvert$ | $< 2.4$ | $< 2.4$ | $< 2.4$ |
| $\Delta R(\ell j)$ (AK8) | $> 0.8$ | $> 0.8$ | – |
| $\Delta R(\ell j)$ (AK4) | $> 0.4$ | – | – |
| $p_T(\ell^+\ell^-)$( GeV) | $> 100$ | $> 200$ | $> 170$ |
| $p_T(V_{had})$( GeV) | $> 100$ | $> 200$ | $> 200$ |
| $\tau_{21}$ | – | $< 0.45$ | Low Purity [0.45,0.75] |
| | | | + High Purity [$< 0.45$] |
| b tagging | Untagged | Untagged | – |
| | + Tagged[CSVL + CSVM] | + Tagged[CSVM] | – |

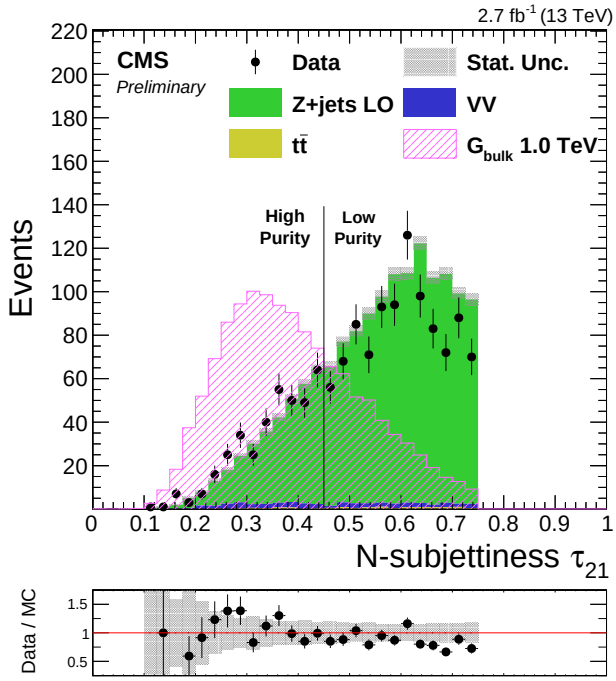


Figure 4: Distribution of $\tau_{21}$ for data and simulated backgrounds in the high mass search; a simulated signal corresponding to a bulk graviton of 1 TeV is also shown. Simulated backgrounds are displayed as stacked histograms normalised to luminosity (2.7 fb$^{-1}$). The boundary at $\tau_{21} = 0.45$ separates the high purity and low purity categories.

# 5 Background Estimation

Most events in the selected ZV candidate sample come from Z+jets production. A real Z boson decaying into $\ell\ell$ and a high $p_{\mathrm{T}}$ merged jet or dijet pair may easily yield a signal-like combination, even though the jets do not stem from a boson decay. Subdominant backgrounds include standard model $\mathrm{t\bar{t}}$, single top, WW, WZ and ZZ production; the latter two are irreducible backgrounds when the diboson pair decays in the $\ell^+\ell^-$ $\mathrm{q\bar{q}}$ channel.

## 5.1 Background Estimation in the Low Mass Search

For the low mass search, the Drell-Yan background is characterised using an inclusive Z+jets sample produced at NLO and a larger set of exclusive Z+jets samples produced at LO. Multiplicative factors, k-factors, are used to correct the expected yield of each exclusive LO sample to that of the NLO in the relevant HT range. It has been observed that at low $p_{\mathrm{T}}$ the LO, even with k-factors applied, gives a poor description of many important variables. Therefore the NLO is used at low $p_{\mathrm{T}}$, where the number of MC events is competitive with respect to the LO and the description of the boson kimematics is superior. The use of the LO is then limited to high $p_{\mathrm{T}}$ where it performs well and is statistically superior to the NLO. The transition points from the NLO to LO are based on the $p_{\mathrm{T}}$ of the dilepton system and occur at values of $p_T(\ell\ell) = 300$ GeV and 200 GeV for the boosted and resolved categories respectively. It has been observed that this NLO+LO sample performs as well as the pure NLO MC, but with the added benefit of an increased number of MC events in the tails of the distributions. However the $m_{\mathrm{ZV}}$ distributions in the b-tagged categories are poorly modelled due to the limited number of Z+jets MC events. It has been observed that, within MC errors, the Z+jets mass shape does not depend on b tagging. Therefore the reconstructed $m_{\mathrm{ZV}}$ shape before b-tag categorisation is used to describe the Z+jets shape in the tagged category.

In the final fit to the data, the normalization of the Z+jets background in the signal region is constrained by the observed yield in the sideband, after subtracting the diboson and $\mathrm{t\bar{t}}$ backgrounds. This procedure is applied independently to each hadronic V-type, lepton flavour and b tagging category. The statistical uncertainties associated with the predicted normalisations of the Z+jets MC distributions are taken into account in the fit. For the shape of the Z+jets $m_{\mathrm{ZV}}$ distribution in the signal region of every category, the NLO+LO MC prediction is taken as a baseline shape. On top of this baseline shape, a family of linear correction functions $\mathrm{Corr}(M_{\mathrm{X}}, s) = 1 + s(M_{\mathrm{X}} - 500)/500$, with one such function defined by the slope parameter $s$, is considered. The choice of this linear correction is motivated by data-MC comparisons in the sideband $m_{\mathrm{ZV}}$ distributions. Fig. 5 displays fits to the sideband $m_{\mathrm{ZV}}$ distributions where the slope parameter $s$ is allowed to float free and is constrained by the observed shapes in data. The two sigma uncertainties on the fitted linear correction functions are observed to cover the residual shape differences in the sideband. In the fit to the signal region data, for every category, the sideband constrained slope parameter $s$ is treated as a Z+jets shape systematic effect, allowing the background shape to be corrected to that of the data.

Dilepton backgrounds that do not contain a real leptonic Z decay are estimated from the data using e$\mu$ events passing the same cuts as the signal [1]. This method accounts for $\mathrm{t\bar{t}}$ production, WW + jets, $\mathrm{Z} \to \tau^+\tau^-$ + jets, single top and fakes. The e$\mu$ vs. $ee + \mu\mu$ symmetry has been verified using a top enriched control sample. Also the e$\mu$ $m_{\mathrm{ZV}}$ distribution has been compared to the lepton flavour symmetric background MC prediction and good agreement has been observed. The contamination from this $t + X$ background represents 1% and 5% of the total background in the untagged and tagged categories of the resolved analysis, while the boosted analyis has a contamination of 1% and 3% ($1 \pm 1$ events) in the untagged and tagged

categories.

The diboson production background (ZZ and WZ, with $Z \to \ell\ell$) is estimated directly from simulation. The contamination from this source represents 4% and 7% of the total background in the untagged and tagged categories of the resolved analysis, while the boosted analyis has a contamination of about 11% and 12% in the untagged and tagged categories.

The background estimates derived from a fit to the sideband region, and the expected signal yields, are compared to the observed number of events in Table 3. The signal region $m_{ZV}$ distributions for the boosted and resolved categories after fitting the signal and sideband regions are depicted in Fig. 6 and 7.

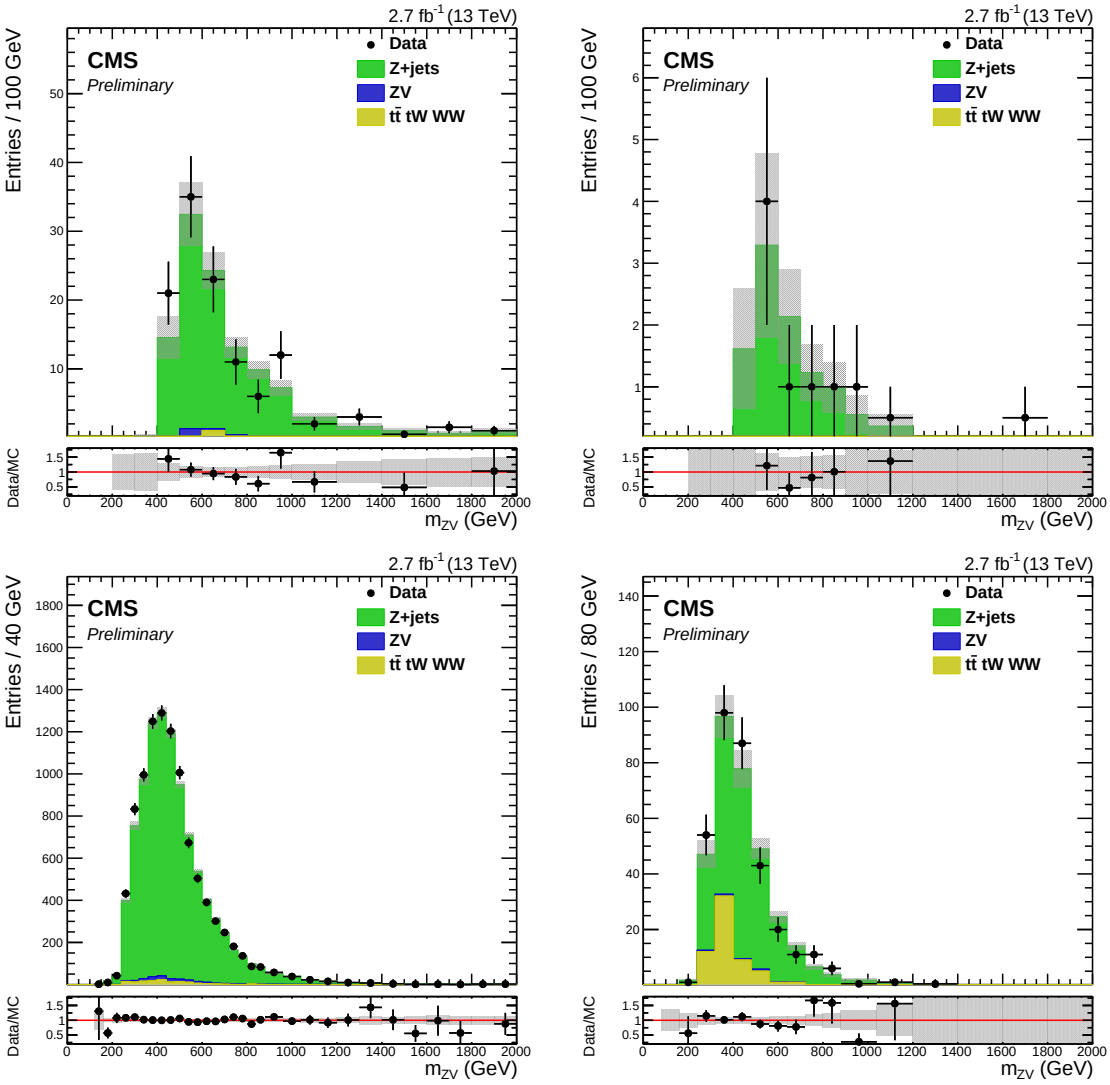


Figure 5: The sideband $m_{ZV}$ distributions for the boosted V (top), resolved V (bottom), untagged (left) and tagged (right) categories of the low mass search after fitting the sideband data alone. The points show the data while the filled histograms show the background contributions. The grey band indicates the statistical and postfit systematic uncertainties on the normalization and shape of the background.

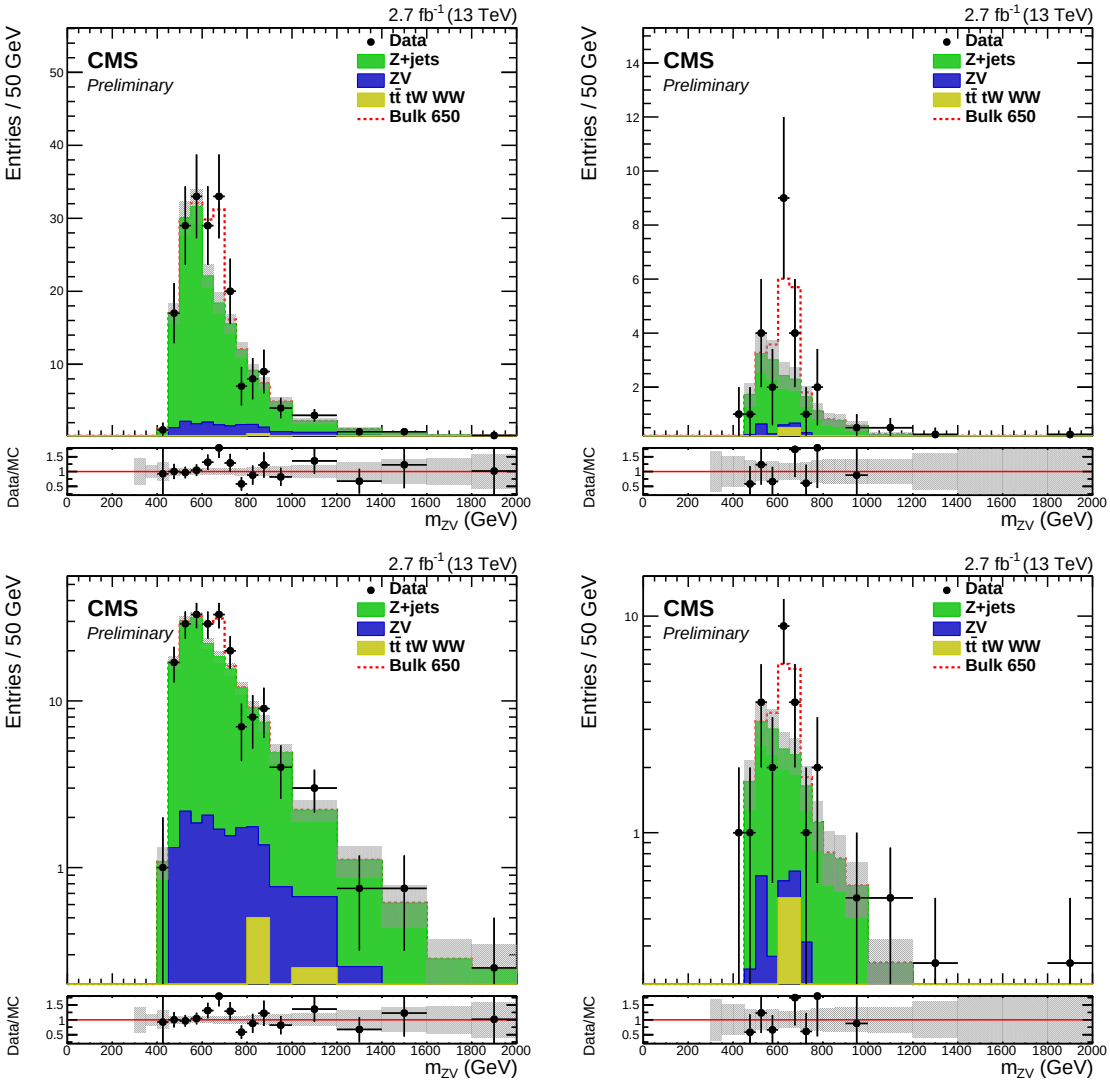


Figure 6: The signal region $m_{ZV}$ distributions for the boosted V-untagged (left) and V-tagged (right) categories of the low mass search after fitting the signal and sideband regions using a signal plus background model. The points show the data while the filled histograms show the background contributions. The signal is represented by the red dashed histogram. The grey band indicates the statistical and systematic uncertainties on the normalization and shape of the background.

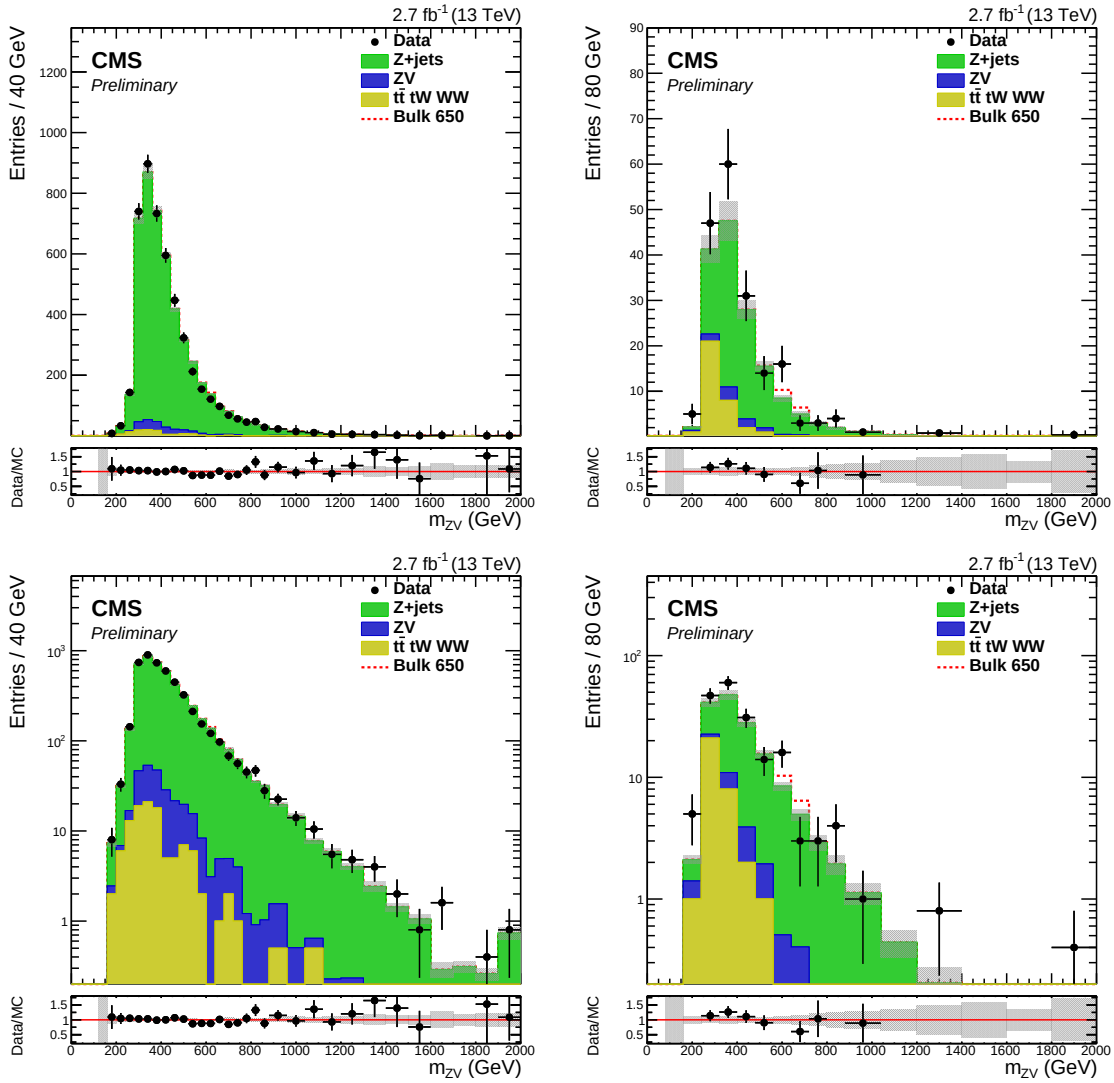


Figure 7: The signal region $m_{ZV}$ distributions for the resolved V-untagged (left) and V-tagged (right) categories of the low mass search after fitting the signal and sideband regions using a signal plus background model. The points show the data while the histograms show the background contributions. The signal is represented by the red dashed histogram. The grey band indicates the statistical and systematic uncertainties on the normalization and shape of the background.

Table 3: Predicted and observed yields with 2.7 $fb^{-1}$ of data for the resolved and boosted categories of the low mass search. The signal and sideband predicted numbers are derived from the fit to the sideband mass distributions alone.

| | Boosted untagged | Boosted tagged | Resolved untagged | Resolved tagged |
|---|---|---|---|---|
| SR predicted | $179 \pm 18$ | $18 \pm 5$ | $2334 \pm 42$ | $65 \pm 6$ |
| SR observed | 194 | 26 | 2327 | 75 |
| SB predicted | $98 \pm 11$ | $9 \pm 3$ | $6420 \pm 93$ | $182 \pm 16$ |
| SB observed | 98 | 9 | 6420 | 182 |
| $m_{ZV}$ (GeV) | RS graviton signal predicted | | | |
| 600 | 662.0 | 201.6 | 1364.2 | 231.1 |
| 800 | 293.0 | 89.8 | 300.2 | 54.9 |
| 1000 | 125.7 | 35.6 | 91.7 | 15.9 |
| 1200 | 47.3 | 15.0 | 30.6 | 4.92 |
| 1400 | 20.2 | 6.19 | 10.4 | 1.63 |
| $m_{ZV}$ (GeV) | Bulk graviton signal predicted | | | |
| 600 | 14.8 | 5.09 | 15.6 | 3.08 |
| 800 | 4.11 | 1.34 | 2.14 | 0.44 |
| 1000 | 1.22 | 0.40 | 0.47 | 0.08 |
| 1200 | 0.41 | 0.13 | 0.10 | 0.02 |
| 1400 | 0.13 | 0.05 | 0.03 | 0.003 |

## 5.2 Background Estimation in High Mass Search

The high mass search estimates the background in the signal region ($65 < m_J < 105$ GeV), using the following inputs:

- Data in the sideband: $(20 < m_J < 65 \text{ GeV}) \cup (135 < m_J < 220 \text{ GeV})$ ,
- MC dominant (Z+jets) in the full region: $20 < m_J < 220$ GeV,
- MC subdominant (VV and $t\bar{t}$) in the full region: $20 < m_J < 220$ GeV.

To avoid bias in the prediction of the background, the actual data in the signal region is kept blind. Therefore, the background prediction in the signal region is derived by looking only at data in the sidebands of the pruned mass distribution. Fig. 8 shows the pruned mass distribution in data and the parametric model made out of two components. The dominant component accounts for the Z+jets background, and the subdominant component corresponds to the diboson VV and $t\bar{t}$ backgrounds. The parametric model consists of an error function multiplied by an exponential (ErfExp); in the low purity categories a gaussian on top of the ErfExp is used for modelling the subdominant backgrounds. The description of the pruned mass with the parametric model ErfExp was tested in simulation and compared against alternative functions. To account for the mismodelling of the pruned mass, an uncertainty was added as a systematic error that ranges between 28-42%. The expected number of events in the signal region is given in Table 4; the result is reported for the categories muon low purity (MLP), electron low purity (ELP), muon high purity (MHP), and electron high purity (EHP). The estimated background is reported in the format A $\oplus$ B, where A and B represent the dominant and subdominant components, respectively.

Table 4: Background estimation for the high mass search obtained by the integral of the parametric model in the signal region ($65 < m_J < 105$ GeV). The estimated background is reported in the format A $\oplus$ B, where A and B represent the dominant and subdominant components, respectively. The systematic uncertainty due to mismodelling of the pruned mass ranges between 28-42%.

| Category | Parametric model | | | Estimated background | | | Syst. unc. | Data in SR |
|---|---|---|---|---|---|---|---|---|
| MLP | (ErfExp) | $\oplus$ | (ErfExp + Gaussian) | $(150 \pm 11)$ | $\oplus$ | $(7.4 \pm 3)$ | 32 % | 157 |
| ELP | (ErfExp) | $\oplus$ | (ErfExp + Gaussian) | $(105 \pm 10$ | $\oplus$ | $(5 \pm 2)$ | 34 % | 116 |
| MHP | (ErfExp) | $\oplus$ | (ErfExp) | $(88 \pm 8)$ | $\oplus$ | $(10 \pm 3)$ | 42 % | 110 |
| EHP | (ErfExp) | $\oplus$ | (ErfExp) | $(53 \pm 7)$ | $\oplus$ | $(9 \pm 3)$ | 28 % | 85 |

A shape correction for the Z+jets component is taking into account through the transfer factor $\alpha(m_{ZV})$. This transfer factor is defined in terms of the simulated $m_{ZV}$ distributions in the sideband and signal regions, as shown in Fig. 9. The transfer factor is applied to the fit of the $m_{ZV}$ distribution in data in the sideband region to obtain the proper shape in the signal region. In addition to the normalization uncertainty due to mismodelling of the pruned mass, the transfer factor $\alpha(m_{ZV})$ introduces shape uncertainties in the final background estimation. These shape uncertainties are quantified in the error bands displayed in Fig. 9.

The $m_{ZV}$ distribution of data in the signal region and the final background estimation are shown in Fig. 10. The decomposition of the parametric model into dominant and subdominant components is also shown. The error band in the parametric model accounts for the shape uncertainties due to the transfer factor, and the uncertainties in the fit of the $m_{ZV}$ distribution in data in the sideband region. The normalization uncertainty (between 28-42 %) due to mismodelling of the pruned mass is considered as a nuisance parameter in the statistical framework that calculates the expected limits. An expanded discussion of systematic uncertainties and the

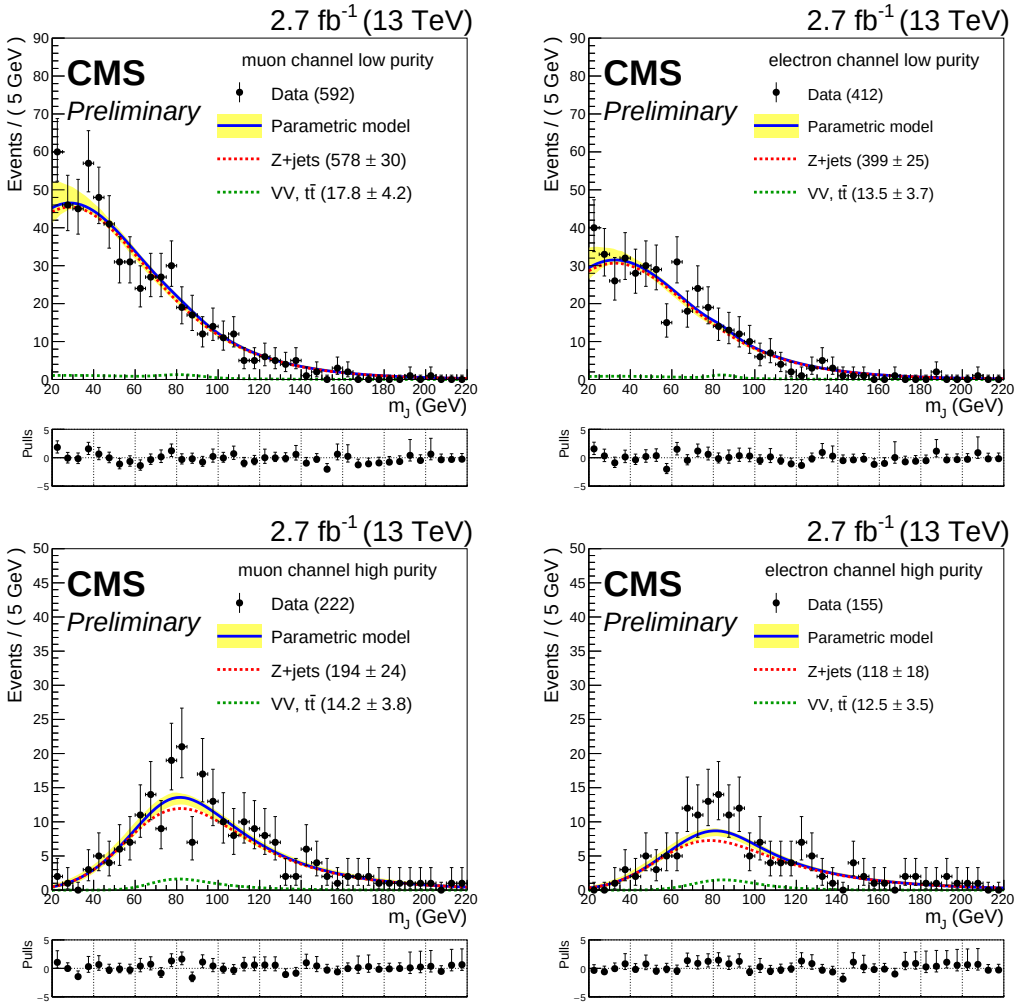


Figure 8: Pruned mass distribution in data and parametric model (blue line) for the high mass search. The dominant (red) and subdominant (green) components are also shown. The yield for each component in the full region ($20 < m_J < 220$ GeV) is in the legend.

statistical treatment is giving in the following sections.

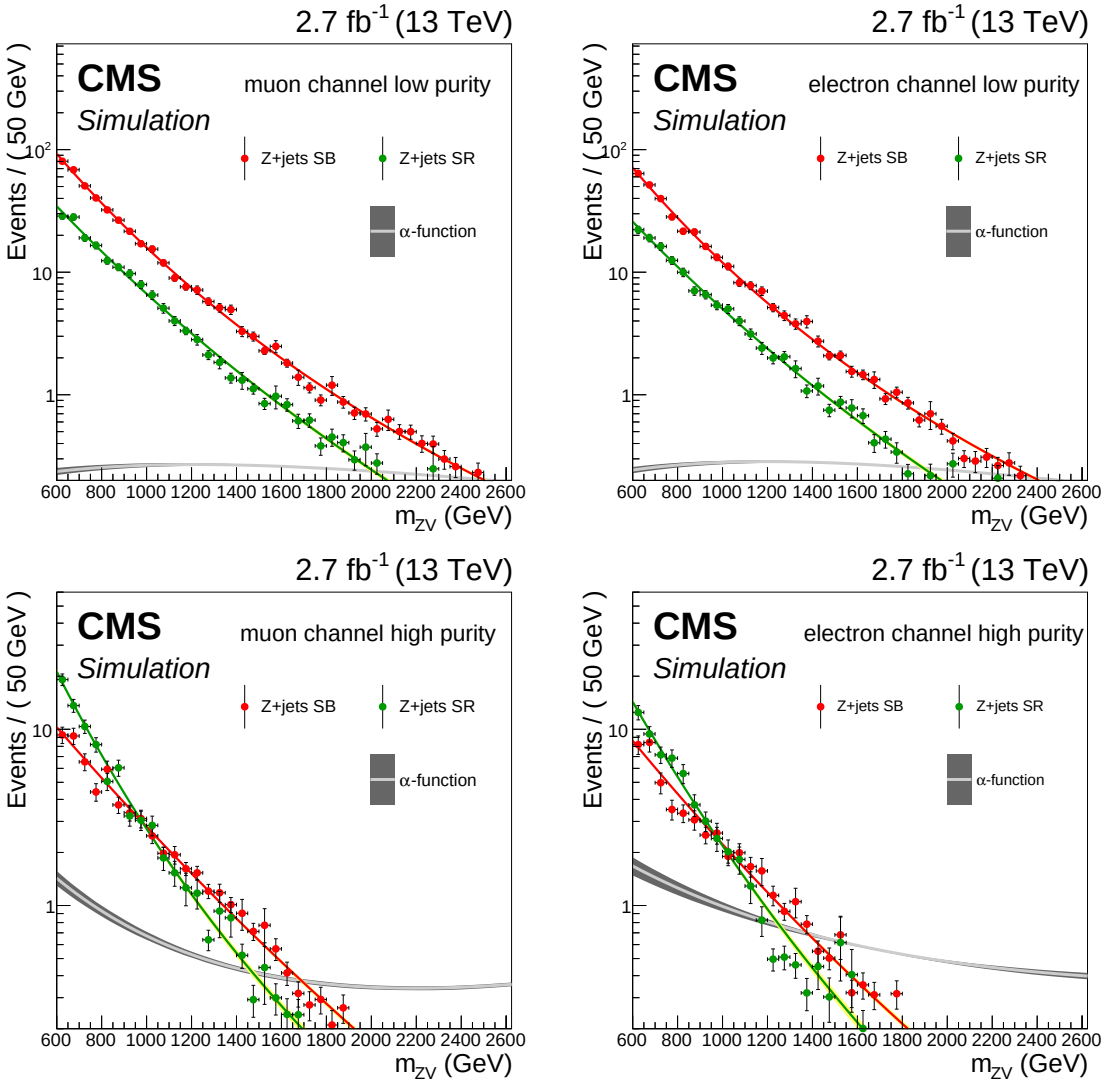


Figure 9: Top: $m_{ZV}$ simulated distributions in the signal (green) and sideband (red) region for the low purity category of the high mass search, for muons (left) and electrons (right). Bottom: $m_{ZV}$ simulated distributions in the signal (green) and sideband (red) region for the high purity category of the high mass search, for muons (left) and electrons (right). The transfer factor $\alpha(m_{ZV})$ is defined by the ratio (Z+jets SR) / (Z+jets SB).

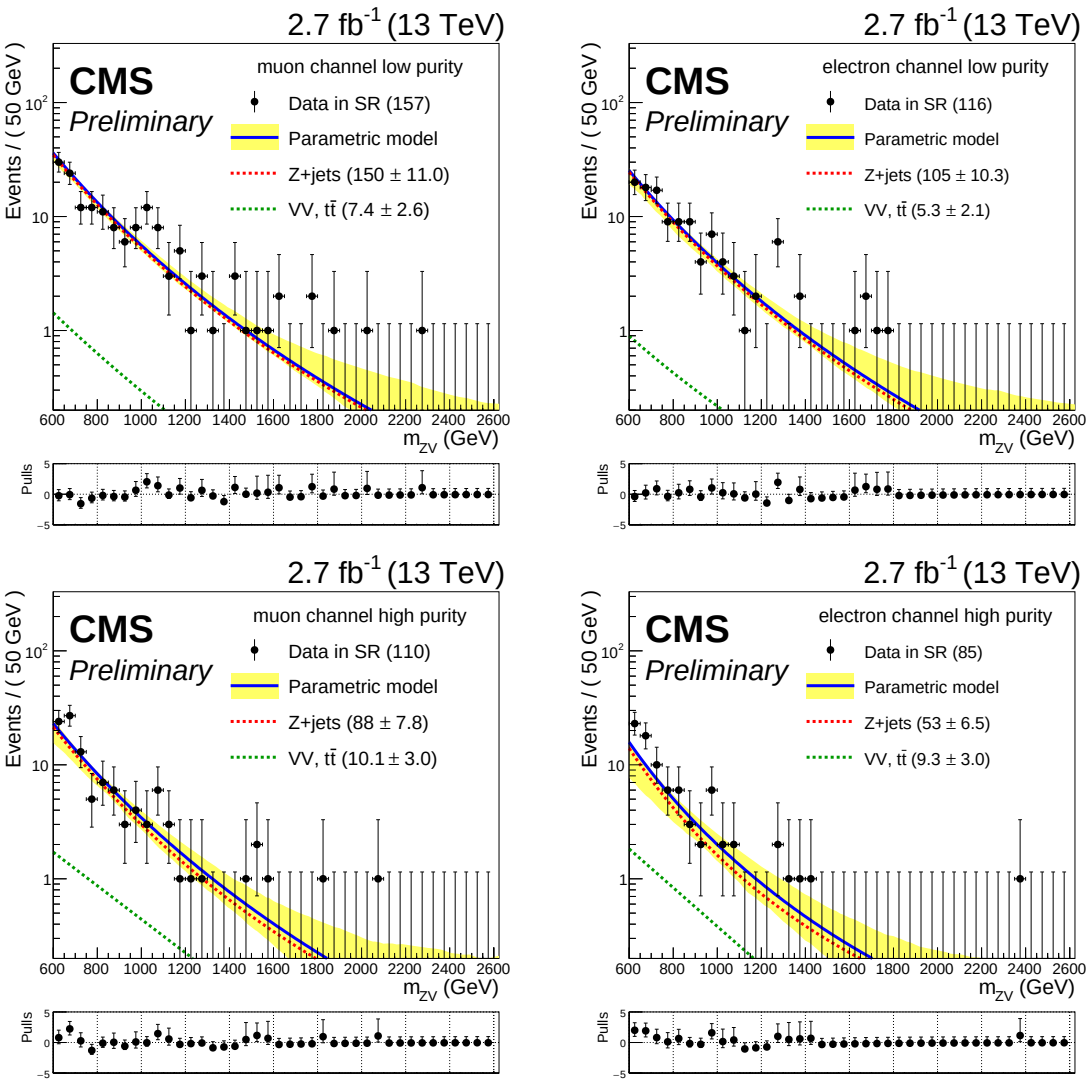


Figure 10: Top: $m_{ZV}$ distributions in the signal region for the low purity category of the high mass search, for muons (left) and electrons (right). Bottom: $m_{ZV}$ distributions in the signal region for the high purity category of the high mass search, for muons (left) and electrons (right).

# 6 Systematic Uncertainties

The systematic uncertainties influence both the normalisation and shape of the backgrounds and signal. We consider effects from leptons (trigger, selection, energy scale), hadrons (jet energy and merged jet pruned mass scale, pileup, heavy quark flavor tagging and mistagging, boosted V tagging), ZV diboson background cross section, $t + X$ background normalisation, signal production mechanism and PDF [22], and LHC luminosity [30].

The dominant uncertainties on the signal selection efficiency arise from uncertainties in the efficiency of tagging the hadronic jet as a V in the low mass and high mass boosted categories, and from uncertainties in the b tagging efficiency in the low mass b-tagged categories. The efficiency of the W tagging selection and its corresponding systematic uncertainty have been measured in data using a sample enriched in semileptonic $t\bar{t}$ events. The b tagging efficiencies and their corresponding systematic uncertainties have been measured in data using samples enriched in beauty decay events [29].

Jet reconstruction uncertainties affect both the signal efficiency and the shape of the $m_{ZV}$ distribution. The momentum and the mass of the reconstructed jets are rescaled according to the uncertainties of the jet energy momentum scale [12]. The selection efficiencies and signal shapes are recalculated on these modified samples, with the resulting changes taken as systematic uncertainties depending on the resonance mass. This effect is responsible for the largest signal shape systematic uncertainty.

Changes in the lepton energy/momentum are propagated through the entire analysis. The relative variation in the number of selected signal events is taken as a systematic uncertainty on signal normalisation. For both lepton flavours, these uncertainties are smaller than 1%.

The systematic uncertainties on the lepton trigger, identification, and isolation efficiencies are derived using tag-and-probe analysis of $Z \to \ell\ell$ events. In the low mass analysis, an uncertainty of 1% is assigned to the trigger efficiency for both lepton flavours; an uncertainty of 1.1% (1.7%) is assigned to the lepton selection efficiency for muons (electrons). In the high mass analysis a systematic uncertainty of 10% (2.5%) is assigned to muons (electrons).

The uncertainty on the integrated luminosity [30] of the data sample (2.7%) introduces an uncertainty in the number of signal events passing the final selection. This uncertainty is fully correlated in all channels. The uncertainty due to parton distribution functions is of the order of 1%.

Signal normalisation systematics in the low mass and high mass analyses are summarised in Table 5. In the low mass analysis, the normalisation systematic errors of the RS and bulk graviton models are found to be equal within MC statistical errors.

Background normalisation systematics in the low mass analysis are summarised in Table 6. To account for background shape systematic effects not explicitly evaluated, we compare data and simulation in the sideband region, and the residual shape difference is treated as another uncertainty; this is the dominant background shape systematic of the low mass analysis.

The background normalisation uncertainties in the high mass analysis are estimated from the covariance matrix of the fit to the data of the $m_{ZV}$ distribution in the sideband region and the uncertainties of the modelling of the $\alpha(m_{ZV})$ ratio, which depend on the number of events in data and simulation, respectively.

Table 5: Summary of systematic uncertainties on the signal normalisation in the low mass resolved, low mass boosted, and high mass analyses.

| Source | Resolved | Boosted | High mass | Comment |
|---|---|---|---|---|
| Luminosity | | 2.7% | | Same for all analyses |
| Electron trigger & ID | 2.0% | 2.0% | 2.5% | Tag-and-probe study |
| Muon trigger & ID | 1.5% | 1.5% | 10% | Tag-and-probe study |
| Electron energy scale | 0.1% | 0.2% | 0.5% | |
| Muon momentum scale | 0.1% | 0.1-0.5% | - | |
| Jet reconstruction | 0.3% | 0.3% | 1% | JES / pruned mass scale, correlated among categories |
| $b$ tagging efficiency SF untagged | 1.4% | 0.4% | - | (Anti)correlated among categories |
| $b$ tagging efficiency SF tagged | 9% | 7% | - | |
| Mistag SF untagged | 0.3% | 0.1% | - | |
| Mistag SF tagged | 0.7% | 0.7% | - | |
| Boosted V tagging | - | 7% | 7% (HP)<br>26% (LP) | Merged jet only |
| Pileup | 0.4% | 0.7% | - | Correlated among categories |
| Production mechanism (PDF) | 1% | 1% | - | PDF4LHC, acceptance only |

Table 6: Summary of systematic uncertainties on the background normalisation in the low mass resolved and boosted analyses.

| Source | resolved | boosted |
|---|---|---|
| Luminosity | 2.7% | |
| Electron trigger & ID | 2.0% | |
| Muon trigger & ID | 1.5% | |
| Electron energy scale | 0.7% | 1.1% |
| Muon momentum scale | 0.2% | 0.3% |
| Jet reconstruction | 3% | 0.7% |
| $b$ tagging efficiency SF untagged | $\pm 0.3\%$ | $\pm 0.2\%$ |
| $b$ tagging efficiency SF tagged | $\mp 9\%$ | $\mp 6\%$ |
| Mistag SF untagged | $\pm 0.1\%$ | $\mp 0.1\%$ |
| Mistag SF tagged | $\pm 1.7\%$ | $\pm 1.7\%$ |
| Boosted V tagging | - | 7% (ZZ/WZ) |
| Pileup | 0.2% | 0.1% |
| Diboson cross section | 12% | |
| $t + X$ normalisation | 4% ($e\mu$) | |

# 7 Statistical Treatment

The low mass search focuses on resonances with masses between 550 GeV and 1400 GeV, and addresses the case where the hadronically decaying vector boson gives rise to either a pair of hadronic jets (resolved category) or a single merged hadronic jet with substructure (boosted category). The high mass search focuses on masses above 800 GeV using merged jet reconstruction and looser lepton reconstruction criteria to recover the efficiency in the boosted regime. Both the low mass and high mass search strategies provide results in the context of the bulk graviton model, while the RS graviton interpretation is considered only for the low mass search. The selection efficiencies obtained from simulation are presented in Fig. 11.

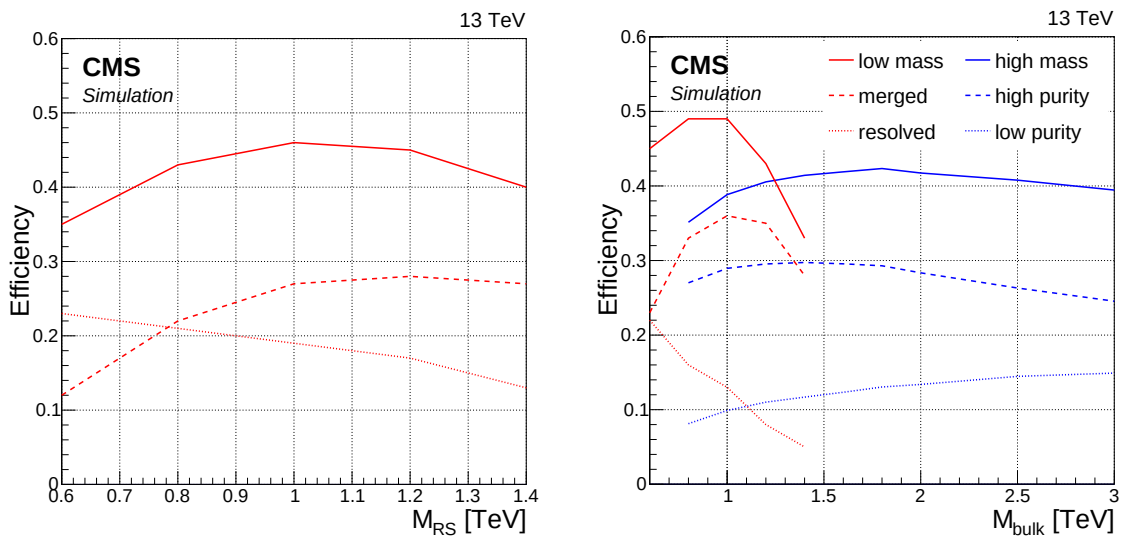


Figure 11: Selection efficiency as function of the RS (left) and bulk (right) graviton mass for the low mass (red solid line) and high mass (blue solid line) searches. The red dashed (dotted) line denotes the efficiency of the low mass search in the boosted (resolved) category; the blue dashed (dotted) line denotes the efficiency of the high mass search in the high purity (low purity) category.

Exclusion limits are extracted using the formalism developed by the CMS and ATLAS Collaborations in the context of the LHC Higgs Combination Group [31]. For each signal mass hypothesis a maximum likelihood fit of the data in each category is performed using a background-only and a signal-plus-background model. A likelihood ratio of the two fits is used as a test statistic for the asymptotic modified frequentist method $CL_s$ [32, 33], with systematic uncertainties incorporated as nuisance parameters. The results are expressed as upper limits on the cross section times branching fraction of the process $X \rightarrow ZZ$ for the RS graviton and bulk graviton models. Fig. 12 shows the observed and expected limits for the full dataset recorded during 2015 at 13 TeV, corresponding to a luminosity of 2.7 $fb^{-1}$. The theoretical NLO graviton production cross-sections and branching ratios, represented by the red lines, have been taken from ref. [34].

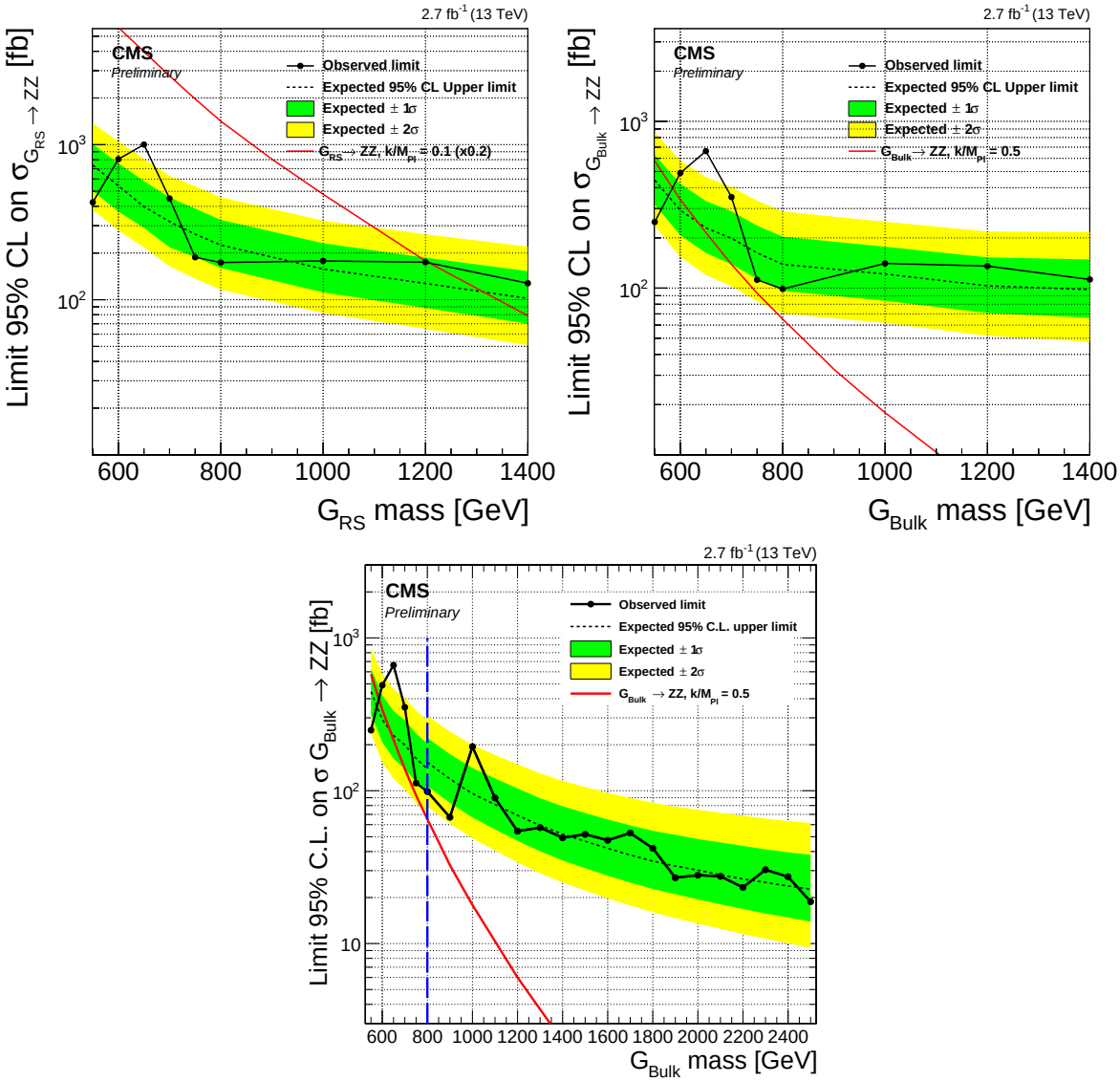


Figure 12: Observed and expected 95% CL upper limits on the RS (top left) and bulk (top right and bottom) graviton production cross section times branching fraction to ZZ obtained by the low mass search (top) and the low and high mass searches combined (bottom). The 68% and 95% ranges of expectation for the background-only model are also shown with green and yellow bands, respectively. The theoretical cross-section for the RS model has been multiplied by a factor of 0.2 for visibility. The blue line in the bottom plot sets the boundary at 800 GeV that separates the low mass and high mass searches.

# 8 Conclusions

A search for diboson resonances in the mass range 550 GeV to 2500 GeV in the semileptonic $X \to ZV \to \ell^+\ell^- q\overline{q}$ final state, where one boson decays hadronically ($V = W, Z$) and the other, a Z boson, decays to two leptons, has been presented. The choice of analysis strategy for a particular signal hypothesis is based on the optimisation of the signal selection efficiency. Data corresponding to an integrated luminosity of 2.7 $fb^{-1}$ of proton-proton collisions at center-of-mass energy of 13 TeV have been analysed.

In the high mass analysis, the results obtained are compatible with the standard model prediction in the explored mass range. Given the low number of events in data and the low cross section of the benchmark model (bulk graviton with $k/M_{Pl} = 0.5$), the present analysis did not reach the sensitivity required to establish exclusion limits on the theoretical cross section of the model under study.

In the low mass search, an excess consistent with the signal at a mass of 650 GeV is observed. The best fit signal strengths are $0.03 \pm 0.01$ times the RS graviton cross section for $k/M_{Pl} = 0.1$, and $2.0 \pm 0.6$ times the bulk graviton cross section for $k/M_{Pl} = 0.5$. The excess has a local p-value (the probability for background alone to give at least as large a signal strength at that mass) of $3.4 \times 10^{-4}$ $(3.4\sigma)$ for the full low mass search (boosted and resolved categories) in the RS interpretation, and $4.0 \times 10^{-5}$ $(3.9\sigma)$ in the bulk graviton case. Taking into account the look-elsewhere effect [35] in the mass range $550 < m_{ZV} < 1400$ GeV, the global p-value is 9.5 times higher in the RS analysis and 10.9 times higher in the bulk graviton analysis, decreasing the global significances to $2.9\sigma$ and $3.5\sigma$, respectively.

# References


[1] CMS Collaboration, "Search for a Higgs Boson in the Mass Range from 145 to 1000 GeV Decaying to a Pair of W or Z Bosons", *JHEP* **10** (2015) 144, doi:10.1007/JHEP10(2015)144, arXiv:1504.00936.

[2] L. Randall and R. Sundrum, "A Large mass hierarchy from a small extra dimension", *Phys. Rev. Lett.* **83** (1999) 3370, doi:10.1103/PhysRevLett.83.3370, arXiv:hep-ph/9905221.

[3] L. Randall and R. Sundrum, "An Alternative to compactification", *Phys. Rev. Lett.* **83** (1999) 4690, doi:10.1103/PhysRevLett.83.4690, arXiv:hep-th/9906064.

[4] K. Agashe, H. Davoudiasl, G. Perez, and A. Soni, "Warped Gravitons at the LHC and Beyond", *Phys. Rev.* **D76** (2007) 036006, doi:10.1103/PhysRevD.76.036006, arXiv:hep-ph/0701186.

[5] A. L. Fitzpatrick, J. Kaplan, L. Randall, and L.-T. Wang, "Searching for the Kaluza-Klein Graviton in Bulk RS Models", *JHEP* **09** (2007) 013, doi:10.1088/1126-6708/2007/09/013, arXiv:hep-ph/0701150.

[6] O. Antipin, D. Atwood, and A. Soni, "Search for RS gravitons via W(L)W(L) decays", *Phys. Lett.* **B666** (2008) 155, doi:10.1016/j.physletb.2008.07.009, arXiv:0711.3175.

[7] CMS Collaboration, "Search for massive resonances decaying into pairs of boosted bosons in semi-leptonic final states at $\sqrt{s} = 8$ TeV", *JHEP* **08** (2014) 174, doi:10.1007/JHEP08(2014)174, arXiv:1405.3447.

[8] CMS Collaboration, "The CMS experiment at the CERN LHC", *JINST* **3** (2008) S08004, doi:10.1088/1748-0221/3/08/S08004.

[9] CMS Collaboration, "Particle-Flow Event Reconstruction in CMS and Performance for Jets, Taus, and MET", Technical Report CMS-PAS-PFT-09-001, CERN, Geneva, 2009.

[10] CMS Collaboration, "Commissioning of the Particle-Flow Reconstruction in Minimum-Bias and Jet Events from pp Collisions at 7 TeV", Technical Report CMS-PAS-PFT-10-002, CERN, Geneva, 2010.

[11] M. Cacciari, G. P. Salam, and G. Soyez, "FastJet User Manual", *Eur. Phys. J.* **C72** (2012) 1896, doi:10.1140/epjc/s10052-012-1896-2, arXiv:1111.6097.

[12] CMS Collaboration, "Determination of Jet Energy Calibration and Transverse Momentum Resolution in CMS", *JINST* **6** (2011) P11002, doi:10.1088/1748-0221/6/11/P11002, arXiv:1107.4277.

[13] CMS Collaboration, "Performance of CMS muon reconstruction in pp collision events at $\sqrt{s} = 7$ TeV", *Journal of Instrumentation* **7** (2012), no. 10, P10002.

[14] CMS Collaboration, "Performance of Electron Reconstruction and Selection with the CMS Detector in Proton-Proton Collisions at s = 8 TeV", *JINST* **10** (2015), no. 06, P06005, doi:10.1088/1748-0221/10/06/P06005, arXiv:1502.02701.

[15] J. Alwall et al., "The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations", *JHEP* **07** (2014) 079, doi:10.1007/JHEP07(2014)079, arXiv:1405.0301.

[16] P. Nason, "A New Method for Combining NLO QCD with Shower Monte Carlo Algorithms", *JHEP* **11** (2004) 040, doi:10.1088/1126-6708/2004/11/040. arXiv:hep-ph/0409146.

[17] S. Frixione, P. Nason, and C. Oleari, "Matching NLO QCD Computations with Parton Shower Simulations: the POWHEG method", *JHEP* **11** (2007) 070, doi:10.1088/1126-6708/2007/11/070. arXiv:0709.2092.

[18] S. Alioli, P. Nason, C. Oleari, and E. Re, "NLO Vector-Boson Production Matched with Shower in POWHEG", *JHEP* **07** (2008) 06, doi:10.1088/1126-6708/2008/07/060. arXiv:0805.4802.

[19] T. Sjöstrand, S. Mrenna and P. Z. Skands, "PYTHIA 6.4 Physics and Manual", *JHEP* **05** (2006) 026, arXiv:hep-ph/0603175.

[20] T. Sjöstrand, S. Mrenna, and P. Z. Skands, "A Brief Introduction to PYTHIA 8.1", *Comput. Phys. Commun.* **178** (2008) 852–867, doi:10.1016/j.cpc.2008.01.036, arXiv:0710.3820.

[21] P. Skands, S. Carrazza, and J. Rojo, "Tuning PYTHIA 8.1: the Monash 2013 Tune", *Eur. Phys. J.* **C74** (2014), no. 8, 3024, doi:10.1140/epjc/s10052-014-3024-y, arXiv:1404.5630.

[22] NNPDF Collaboration, "Parton distributions for the LHC Run II", *JHEP* **04** (2015) 040, doi:10.1007/JHEP04(2015)040, arXiv:1410.8849.

[23] GEANT4 Collaboration, "GEANT4: a simulation toolkit", *Nucl. Instrum. Meth.* **A506** (2003) 250, doi:10.1016/S0168-9002(03)01368-8.

[24] CMS Collaboration, "Measurements of inclusive W and Z cross sections in pp collisions at $\sqrt{s} = 7$ TeV", *JHEP* **01** (2011) 001, doi:10.1007/JHEP01(2011)080, arXiv:1012.2466.

[25] M. Wobisch and T. Wengler, "Hadronization corrections to jet cross-sections in deep inelastic scattering", in *Monte Carlo generators for HERA physics. Proceedings, Workshop, Hamburg, Germany, 1998-1999*.

[26] S. D. Ellis, C. K. Vermilion, and J. R. Walsh, "Techniques for improved heavy particle searches with jet substructure", *Phys. Rev.* **D80** (2009) 051501, doi:10.1103/PhysRevD.80.051501, arXiv:0903.5081.

[27] J. Thaler and K. Van Tilburg, "Maximizing Boosted Top Identification by Minimizing N-subjettiness", *JHEP* **02** (2012) 093, doi:10.1007/JHEP02(2012)093, arXiv:1108.2701.

[28] CMS Collaboration, "Combination of diboson resonance searches at 8 and 13 TeV", Technical Report CMS-PAS-B2G-16-007, CERN, Geneva, 2016.

[29] CMS Collaboration, "Identification of b quark jets at the CMS Experiment in the LHC Run 2", Technical Report CMS-PAS-BTV-15-001, CERN, Geneva, 2016.

[30] CMS Collaboration, "CMS Luminosity Measurement for the 2015 Data Taking Period", Technical Report CMS-PAS-LUM-15-001, CERN, Geneva, 2016.

[31] ATLAS and CMS Collaborations, "Procedure for the LHC Higgs boson search combination in Summer 2011", *ATL-PHYS-PUB-2011-11, CMS NOTE-2011/005* (2011).

[32] T. Junk, "Confidence level computation for combining searches with small statistics", *Nucl.Instrum.Meth.* **A434** (1999) 435, doi:10.1016/S0168-9002(99)00498-2, arXiv:hep-ex/9902006.

[33] A. L. Read, "Presentation of search results: the CLs technique", *J. Phys. G: Nucl. Part. Phys.* **28** (2002).

[34] A. Oliveira, "Gravity particles from Warped Extra Dimensions, a review. Part I - KK Graviton", arXiv:1404.0102.

[35] E. Gross and O. Vitells, "Trial factors or the look elsewhere effect in high energy physics", *Eur. Phys. J.* **C70** (2010) 525, doi:10.1140/epjc/s10052-010-1470-8, arXiv:1005.1891.